

MA2031-Hints to solve problems of Assignment IV

1. (a) Zero vector is not mapped to the zero vector.
(b) T does not preserve vector addition.
(c) T does not preserve vector addition.
2. No.
3. (a) No.
(b) Yes, $T(x, y, z) = (2x - y, x - y, 2x)$
(c) No.
(d) Yes, $T(x, y, z) = (z, \frac{1}{2}(-x + y + z))$
4. If inner product is preserved, then it preserves the length is trivial. Use the polarization identity to prove the other way.
5. (a) Nullity of T is 0, then use rank nullity theorem to find $\text{rank}(T)$.
(b) Nullity of T is 1, then use rank nullity theorem to find $\text{rank}(T)$.
(c) Nullity of T is 0, then use rank nullity theorem to find $\text{rank}(T)$.
6. Space of constant functions.
7. (a) $T(x, y) = (x + y, -x - y)$
(b) Take any two linear transformations from $\mathbb{R}^3$ to $\mathbb{R}^3$ such that nullity of the transformations are zero. (Also think of transformations of nullity 2 or 1)
8. (a) Yes.
(b) Yes. Try to construct such linear transformations.
9. Use the definitions of a linearly independent set, spanning set, basis and linear transformation to prove the three results.
10. Trivial.
11. Use the fact that $R(TS) \subset R(T)$ to prove $\text{rank}(TS) \leq \text{rank}(T)$. Use $N(S) \subset N(TS)$ to prove that $\text{rank}(ST) \leq \text{rank}(S)$.
12. Trivial.
13. Use rank nullity dimension theorem.
14. Define $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ such that $T(x, y) = (x, -x + y, 0)$ and $S : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ such that $S(x, y, z) = (x, x + y)$, then $ST = I$ but $TS \neq I$.